

Anish Kumar

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EDUCATION

Indian Institute of Technology (IIT), Patna	CGPA: 7.34	Aug 2023 – May 2027 (expected)
<i>Bachelor of Technology in Mathematics and Computing</i>		
Relevant Coursework: Data Structures and Algorithms, Number Theory, Machine Learning, Object Oriented Programming, Database Management Systems, Probability and Statistics, Artificial Intelligence, Convex Optimizations, Computer Architecture		

Senior Secondary (CBSE) Percentage: 91.2% **2023**

SKILLS

Core Concepts: Data Structures & Algorithms, Graph & Number Theory, Probability & Statistics, Operating Systems, DBMS, OOPs
Databases & Tools: SQL (PostgreSQL, MySQL), pgvector, Git, GitHub Actions, CI/CD, Docker, Kubernetes, Vercel, Firebase, MongoDB
Systems: Linux, Azure, Shell Scripting, Performance Optimization, Multithreading (basic)
Languages: C++, Python, Java, JavaScript, TypeScript, Go
Frameworks: React, React Native, Next.js, Node.js, FastAPI, Tailwind
Libraries: NumPy, Pandas, OpenCV, XGBoost, Random Forest

EXPERIENCE

Optum Global Solutions (UHG)	Bengaluru, India	June 2026 – Present
<i>Software Engineering Intern</i>		
- Built the Production Update Database Tool on Azure/Kubernetes to automate query execution for 1,000+ teams, optimizing parsing logic to evaluate 5,000+ complex queries in under 0.3s. - Engineered a pgvector and MySQL data ingestion pipeline featuring an Isolation Forest ML model for drift detection, coupled with a real-time notification service to alert admins. - Architected an agentic Retrieval-Augmented Generation (RAG) and LLM pipeline with a custom SQL parser and rules engine to surface plain-language summaries and secure rollback scripts.		

TLE Eliminators	Remote	2025 – Present
<i>Post-Contest Discussion (PCD) Head</i>		
- Led post-contest discussions for a large competitive programming community, recording and publishing 15+ live video breakdowns of complex algorithmic problem solutions. - Demonstrated strong technical communication and organizational skills by translating advanced problem-solving techniques and data structures into highly accessible educational content for diverse audiences.		

ACHIEVEMENTS

- **Candidate Master** (1952) on **Codeforces** - **Profile Link**
- **Ranked 186th globally** in **Meta Hacker Cup 2025**, progressed to Round 3 - **Certificate**
- **Selected for Amazon ML Summer School 2026**
- **Semifinalist** in **Flipkart Grid 8.0** (2026)
- **Secured 6th rank** among 6,000+ submissions in **NK Securities Research Hackathon** for Volatility Curve Prediction - **Certificate**
- **4 Star** (1839 rated) on **CodeChef** - **Profile Link**
- **Global Rank 40** in **CodeChef Starters 219**
- **Achieved rank 337** among **38,278** participants in **Codeforces Round 1027 (Div. 3)**
- **Reached Round 2** in **The Big Code Challenge by Google 2026**
- **Selected for Summer of Bitcoin 2025**; contributed to **ZeusLN Wallet** in Linux environment by implementing and validating on-chain message signing and verification using **yarn test**, **lint**, **tsc**, and **prettier**
- **Ranked top 0.76%** in **JEE Advanced** among 1.1M+ candidates

PROJECTS

OneIITP (App Development)  PlayStore  AppStore  OneAttend	Aug 2024 – Present
- Built and launched IIT Patna's official student app for 2,000+ active users, increasing engagement by 40% and generating 15,000+ reads, 1,500+ downloads, and a 4.9/5 rating. - Developed a QR-based anti-proxy attendance system with auto-refreshing codes, cutting proxy attendance by 90%, and created the OneAttend portal for professor-side attendance tracking and analytics. - Integrated Azure authentication and Google Sheets API to sync real-time bus schedules and campus data, improving convenience for students and staff by 30%.	

Facility Routing Engine  Github  Live Website	Mar 2026
Tech Stack: Next.js, React 19, TypeScript, Zustand, Canvas API, A*/Dijkstra, Vercel - Modeled indoor navigation as a weighted graph, implementing A* and Dijkstra algorithms for optimal pathfinding. - Optimized pathfinding computations by reducing redundant graph traversals, improving query efficiency under dynamic constraints. - Implemented emergency-aware pathfinding for fire, medical, security, smoke, and hazmat scenarios with dynamic rerouting around hazardous zones. - Added constraint-based routing rules such as fire-safe routing, emergency-zone avoidance, and stair preference over lifts during fire scenarios.	

POSITIONS OF RESPONSIBILITY & VOLUNTEER EXPERIENCE

Dev Lead for Cyber Security Problem Statement: <i>ICTC 3.0</i>	Mar 2025 – Apr 2025
–  <i>Deployed Website Solution</i>  <i>Problem Statement</i>	
Overall Tech Coordinator <i>Anwesha, IIT Patna</i>	Apr 2025
Dev & Design Team <i>Inter IIT Cult 2024, IIT Patna</i>	Dec 2024

CERTIFICATIONS

Certificate of Achievement: Murf AI Coding Challenge 2025 Certificate Link	Jun 2025
Google Cloud Computing Foundations: Infrastructure in Google Cloud Certificate Link	Oct 2023